

Diagnosing Reversed and Non-Equivalent Ratios

Answer Key

Grade 6

Quick Answers

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|--|--|
| 1. corrected ratio = $\frac{5}{3}$, — | 6. corrected second number = 12, — |
| 2. = or not =, — | 7. correct comparison = $<$, — |
| 3. correct speed = 40 mph, — | 8. corrected value under 15 = 6, corrected value |
| 4. corrected value = 9, — | under 25 = 10, — |
| 5. correct cups = 3, — | |
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What is verified

0 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 6

6.RP.A.2 – problems 1, 3

6.RP.A.3 – problems 2, 4, 5, 6, 7, 8

Difficulty 1–4 of 5 across 17 tagged checks.

Common wrong answers

1. If they got 0.6: reversed the order, writing oats to milk instead of milk to oats
 3. If they got 0.025: divided hours by miles instead of miles by hours
 4. If they got 10: added 4 to the previous milk value (6) instead of keeping the 2:3 ratio
 5. If they got 48: flipped one side of the proportion, solving $2/8 = 12/x$ instead of $2/8 = x/12$
 8. If they got 8: copied the 8 from the next column instead of scaling the 5:2 ratio
 8. If they got 12: added 4 to the previous bottom entry (8) instead of scaling the ratio
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1. Kim writes the ratio of milk to oats (5 cups milk, 3 cups oats) as 3:5. Correct it and say what she mixed up.

Solution: Milk is named first, so the milk count (5) must come first. The correct ratio of milk to oats is

$\frac{5}{3}$

Diagnosis (instructor judges): Kim reversed the order — she wrote the oats-to-milk ratio. Full credit for saying the quantity named first goes first, so milk to oats is 5 to 3, not 3 to 5 (which is 0.6 as a fraction, the reversed value).

2. Jon says 4:6 and 10:15 are equivalent. Is he right?

Solution: Simplify both:

$$\frac{4}{6} = \frac{2}{3} \qquad \frac{10}{15} = \frac{2}{3}$$

Both reduce to the same simplest form, so Jon is right:

$$\boxed{\frac{4}{6} = \frac{10}{15}}$$

Explanation (instructor judges): Full credit shows both ratios simplified to $\frac{2}{3}$ (or one obtained from the other by multiplying both parts by 2.5).

3. Leo computes the speed of a car that travels 120 miles in 3 hours as $3 \div 120 = 0.025$ miles per hour. Fix it and name the mistake.

Solution: Speed in miles per hour is miles divided by hours: $120 \div 3 = 40$.

$\boxed{40 \text{ mph}}$

Diagnosis (instructor judges): Leo divided hours by miles — his 0.025 is hours per mile, the reversed rate. “Miles per hour” tells you the order: the quantity before “per” goes on top.

4. Rice to milk should be 2:3; the table shows columns (2,3), (4,6), (6,10), (8,12). Find and fix the bad entry.

Solution: Test each column: $3 \div 2 = 1.5$, $6 \div 4 = 1.5$, $10 \div 6 \approx 1.67$, $12 \div 8 = 1.5$. The third column breaks the pattern.

(a) The wrong entry is the 10 (paired with 6 cups of rice).

(b) For 6 cups of rice the multiplier from the base column is 3, so milk should be $3 \times 3 = 9$.

$\boxed{9}$

The planted 10 comes from adding 4 to the previous milk value instead of keeping the ratio.

5. Mia’s proportion for 12 muffins (8 muffins need 2 cups) is $\frac{2}{8} = \frac{12}{x}$, giving $x = 48$. Explain and fix.

Diagnosis (instructor judges): The left fraction is cups over muffins, but the right fraction is muffins over cups — Mia flipped one side. Both sides must keep the same order.

Solution: With matching orders, $\frac{2}{8} = \frac{x}{12}$, so $x = \frac{2}{8} \times 12 = 3$ cups of flour.

$\boxed{3}$

6. Which of 6:8, 9:10, 15:20 is not equivalent to 3:4, and what should the second number be?

Solution: (a) 6:8 simplifies to 3:4 and 15:20 simplifies to 3:4, but 9:10 does not — it comes from *adding* 6 to both parts of 3:4. The odd one out is 9:10.

(b) Keeping the 9: since $9 = 3 \times 3$, the second number must be 4×3 .

12

7. Raj writes $\frac{7}{3} > \frac{4}{9}$ to compare punch A (3 syrup : 7 water) with punch B (4 syrup : 9 water). What went wrong, and what is the fair comparison?

Diagnosis (instructor judges): Raj compared water-per-syrup for A against syrup-per-water for B — two different kinds of ratios, so the comparison means nothing. Full credit for naming the mismatched orders.

Solution: Using syrup per water for both: $\frac{3}{7} \approx 0.429$ and $\frac{4}{9} \approx 0.444$, so

$$\frac{3}{7} < \frac{4}{9}$$

Punch B is actually the stronger one.

8. The 5:2 table shows top 10, 15, 20, 25 over bottom 4, 8, 8, 12. Fix the two wrong bottom entries and explain the test.

Solution: Each column must simplify to $\frac{5}{2} = 2.5$. Test: $10/4 = 2.5$ ✓, $15/8 = 1.875$ (×), $20/8 = 2.5$ ✓, $25/12 \approx 2.08$ (×).

(a) Under 15: $15 \div 2.5 = 6$, so the bottom entry is

6

(b) Under 25: $25 \div 2.5 = 10$, so the bottom entry is

10

(c) **Explanation (instructor judges):** Full credit describes a per-column test — divide top by bottom and check for $\frac{5}{2}$, or simplify each column ratio. The planted 8 copies the neighbouring column, and the planted 12 adds 4 to the previous entry instead of scaling.